

Interactivity as Embodied Engagement in *Lamentation: Dancing the Archive*

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Figure 1: Performance demo and interaction testing for *Lamentation: Dancing the Archive*, featuring Lloyd Knight of the Martha Graham Dance Company, February 2025 at the Barnard Movement Lab.

Abstract

This paper describes the creative process for *Lamentation: Dancing the Archive*, which transforms Martha Graham's iconic 1930

solo, *Lamentation*, into an interactive installation that enables audiences to experience this historic work through embodied engagement. Using volumetric video capture and gesture-based interaction, the project allows viewers to control and explore the performance through their own movements, creating an intimate physical relationship with archival choreographic material. This work demonstrates how interactive technology can unlock embodied connections with dance archives, proposing that engaging users' kinesthetic capacities alongside visual perception can yield meaningful experiences with historical works and expand traditional approaches to digital humanities and archival practice.

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CCS Concepts

• **Human-centered computing** → **Visualization design and evaluation methods**; • **Computer systems organization** → *Real-time system architecture*; • **Computing methodologies** → *Motion capture*; *Mixed / augmented reality*; **Volumetric models**; **Image manipulation**.

Keywords

Performance; Archive; Interaction Design; Volumetric Film; Performance Studies; Dance History

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1 Introduction

Martha Graham once described the experience of performing her work *Lamentation* as “stretching inside of your own skin” [7]. This visceral image of emotion, not expressed but embodied, captures the haunting power of the solo. Premiering in 1930 and set to music by the Hungarian composer Zoltán Kodály, *Lamentation* is performed almost entirely from a seated position. The dancer, encased in a tube of purple jersey, contorts and contracts within the fabric. Rather than portray a character, the dancer embodies a condition: grief, made manifest.

Within that tube, the shape of the body and the nature of the movement align, and the physical and conceptual lines of the work become visible and recognizable. The audience witnesses the long diagonal stretch, the curve of sorrow, and the sculptural silhouette frozen mid-breath, and the dance begins to resemble something timeless, composed, and beautiful. But for the dancer, grief is not compositionally contained; it exists in the vibration of the performance. The dancer presses against the limits of form, continually stretching and pushing beyond a representation of grief. Instead, the dancer wrestles with it. The jersey resists breath, swallows movement, and refuses stillness. Every angle is earned, and every contraction represents survival. Yet, from the outside, much of this labor disappears. The audience sees clarity, elegance, and minimalism. Over time, *Lamentation* has become fixed in its own iconic status—respected, studied, but distanced. The viewers’ lack of embodied experience has kept the solo within the realm of abstraction and aestheticism.

How do we help the audience comprehend what Graham meant when she said, “to stretch inside of your own skin?” Can technology offer a new means of communication, one that enables audiences to feel more, not less? To move more, not less? This question became the foundation for the exploration engaged by the co-authors of this paper: can we offer an experiential pathway for contemporary audiences to dance with history, with an archive?

Lamentation: Dancing the Archive is the culmination of this exploration, inviting audiences into an intimate, physical relationship with a specific historic choreography. By reimagining Graham’s iconic 1930 solo through gesture-based interaction with volumetric

video and motion-tracking technology, the project transforms passive spectatorship into an active, embodied encounter. The viewer is invited not just to observe *Lamentation*, but to immerse themselves in its atmosphere; to feel its tremble in shared time.



Figure 2: Early performance testing with co-author Xin Ying at SVstudios in Los Angeles, demonstrating 3D image scaling as hands widen.

2 Interactivity and Embodiment

Lamentation: Dancing the Archive is an interactive installation in which audience members are invited to dance with a volumetric film of *Lamentation*. The installation tracks the hand motions of the viewer using MediaPipe computer vision software in TouchDesigner [4, 6], whose float values are sent to Unity [9] via OSC to mediate control of a 3D environment. The viewer’s hand motions and gestures impact the scale of the 3D film, as well as the position of the camera around it and the lighting inside of the 3D scene, transforming the visual landscape of the work as the user moves alongside it, as seen in Figure 2.

This participatory structure enables a new kind of connection to archival material—one that is felt through the body rather than mediated solely through text or image. Audiences engage with the work in a spatial/somatic relationship, generating an experiential understanding of the choreographic structures. In doing so, they enter into a dialogue with both the past and the present, creating an evolving archive that honors tradition while expanding its boundaries. The installation becomes a site of co-creation where viewers’ bodies serve as both interface and instrument.

The project draws inspiration from the tradition of expanded cinema, a term used to describe film, video, multi-media performance, or immersive environments that push the boundaries of cinema and reject the traditional one-way relationship between audience and screen [10]. In *Lamentation: Dancing the Archive*, the screen does not simply transmit; it listens and responds. This endeavor finds a community of practice alongside practitioners such as Kate Elswit and Harmony Bench’s data-driven visualizations in “Dunham’s Data: Digital Methods for Dance Historical Inquiry and Radical Accounting” [3] and Grisha Coleman’s *The Movement Undercommons* [5]. Such projects seek to recontextualize and reimagine dance history and archival practices through informational, interactive, and site-specific choreographic and presentational logics.

This work further echoes the spirit of exhibitions such as “Dancing the Algorithm” at Jacob’s Pillow Dance Festival, curated by this paper’s co-author Katherine H. Fisher, and where this project premiered in July 2025. In this exhibit, audiences are invited to “not just use technology—we wear it, speak to it, and even dance with it. It anticipates our desires, becoming an interlocutor in our becoming.” Interactivity here is not just a technical feature but a philosophical orientation; a recognition that technology, rather than being external and imposing, is increasingly intimate and entangled with a sense of presence. In this context, technology is not merely a tool of control but a site of connection and embodied meaning-making.

3 Technical Implementation

Utilizing the Evercoast volumetric system with 22 Azure Kinect cameras, we captured two principal dancers from the Martha Graham Dance Company and reconstructed their performance as a high-fidelity 3D video using Depthkit’s reconstruction pipeline [8]. Post-processing was completed in Adobe AfterEffects [1], with final visualization and interaction implemented in Unity and TouchDesigner.

What distinguishes our pipeline is its integration of embodied interaction through real-time motion tracking and control, as demonstrated in Figure 3. A web camera captures the viewer’s body position as they sit on the installation bench, utilizing MediaPipe and Open Sound Control (OSC) to translate gestures into parameters for playback control, including starting, stopping, scaling, lighting, and rotating the viewpoint around the volumetric video. This transforms the video into a responsive, immersive system, allowing viewers to “dance with the archive” and explore the performance from spatial angles that are unavailable to audiences in the stage performance and in traditionally filmed versions of the work, which rely on frontal orientation.

A key challenge we encountered was the system’s difficulty in reconstructing delicate fabrics. The lidar sensors struggled with translucent or soft materials. For *Lamentation*, this issue becomes particularly salient, as the choreography relies on the stretching, folding, and manipulation of a soft jersey tube. In our installation, the audience sees mesh dropout and visible holes. Artistically, this has a poetic resonance, offering the work a digitized aesthetic and gesturing to the incomplete nature of any archival or reconstructed work.

4 Usability and Engagement

To evaluate usability and engagement, we conducted two rounds of user testing with a broad demographic, including artists, dancers, students, and general art audiences. Delays in system responsiveness were broadly interpreted as technical failures, highlighting the importance of real-time feedback in interaction design and prompting technical efforts towards responsiveness and immediacy [2]. Early experimentation also demonstrated that relying exclusively on audience curiosity for intuitive, embodied exploration yielded disengagement and disinterest, even in populations such as dancers with a high level of athletic and embodied knowledge. Put simply: in order to deeply engage with the work, users needed to adequately learn the choreography that activates it, and to have

that choreography yield an immediate and apparent installation response.

We tested a number of instructional frameworks, including written instructions delivered as wall text; written instructions delivered on-screen; a graphical ‘key’ with drawings of available gestures; and a demonstration video performed by a company dancer. The most reliably effective method of instruction was the demonstration video, which was accompanied by auditory and written cues directing participants how to perform the inciting gestures. Participants struggled to rehearse or remember gestures that were explained to them in writing or with representative images; however, the ability to see and practice the gestures significantly increased both comprehension and the time spent interacting with the piece. Participants noted the ability to stop, rotate, and examine the dancer in motion from all angles as highly engaging and emotionally resonant, often spending several minutes manipulating and exploring the work. One participant even started imitating the filmed performer, enacting the choreography of *Lamentation* as the image moved along with them. Afterwards, this participant reported that “my body needed that motion,” indicating a deep emotional connection with the material and a sense of catharsis inspired by the work.

5 Conclusion

Lamentation: Dancing the Archive uses novel technical methods, including real-time motion tracking and volumetric video, to deepen audience engagement with a specific historical work. The work ultimately proposes that engaging a user’s kinesthetic and spatial (as well as visual) capacities can yield impactful and resonant emotional experiences in broad arts audiences. Inspired by the connection audiences feel to this work, enabled by the union of embodied action and technical capacity, co-authors of this paper are exploring further archival projects involving volumetric video, AI, and interactive media. Specifically, one technical issue being further explored is the challenge of fabric capture in volumetric 3D, which is guiding future experiments in stitching and mesh refinement techniques, including the use of Gaussian Splatting.

Lamentation: Dancing the Archive moves beyond the reenactment of a masterpiece. As an artistic and technical endeavor, this project’s creative team seeks instead to unlock the shared emotionality within it—to make tangible and embodied Graham’s century-old exploration of grief. This work speaks beyond time, and beyond the body of a single dancer, to a multiplicity of individuals who continue to engage with this work and Martha Graham’s legacy.

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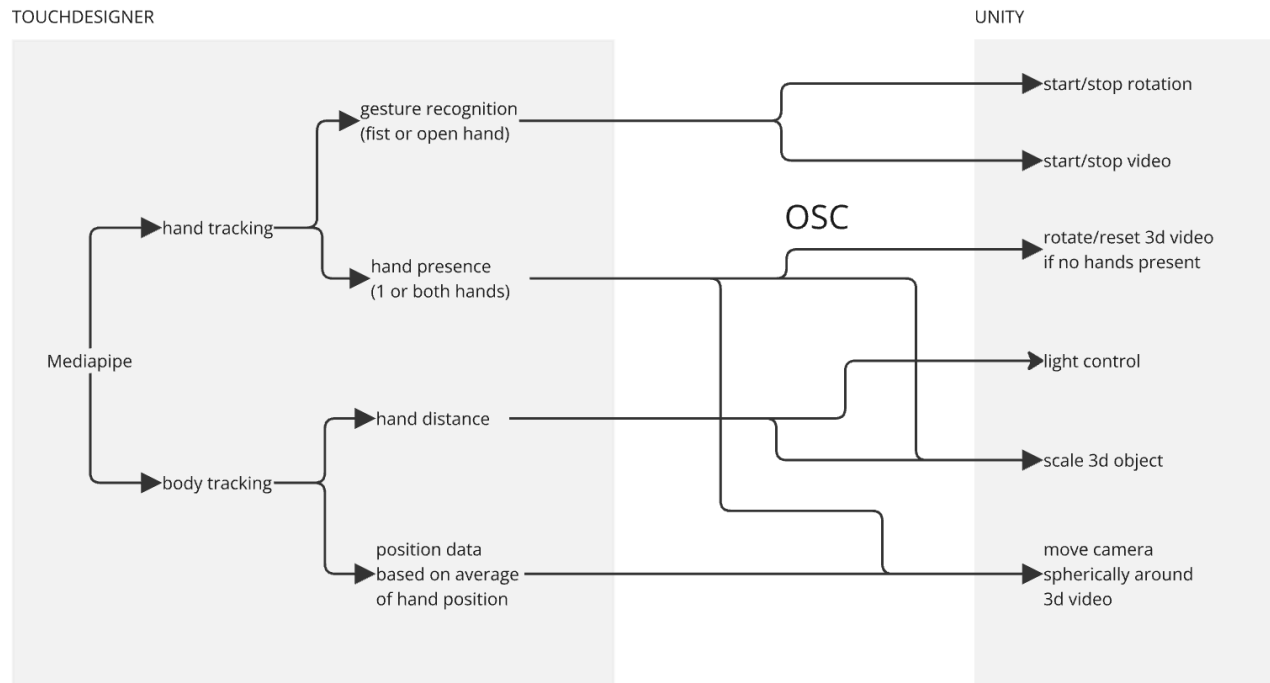


Figure 3: Technical implementation diagram, showing flow of information from TouchDesigner to Unity 3D environment.

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